

Preventing new-borns' brain damage, the low-cost way

Sometimes, conditions like umbilical cord entanglement and anaemia can deprive babies of oxygen just before birth, causing brain damage and cerebral palsy. The best way to prevent this is 'therapeutic hypothermia' or cooling of the baby for an extended period of time, under safe conditions. Well-equipped hospitals have incubator-like cooling devices, which maintain a specific temperature till the baby recovers, but these cost around \$12,000 and are not affordable by all.

Recently, students at John Hopkins University, Maryland, USA, have developed a low-cost device called Cooling Cure, for reducing the temperature of babies at a cost of just \$40.

Inside: The Cooling Cure system can reduce a new-born's temperature by about six degrees for three consecutive days. The cooler is made of a clay pot and a plastic-lined basket, separated by a layer of sand and urea-based powder, similar to the cooling powder used commonly in instant cold-packs. When water is added to the sand and powder mixture, the resulting chemical reaction draws heat away from the upper basket, which cradles the child. A battery-powered microprocessor and sensors track the child's internal and skin temperatures. A green light indicates proper temperature, red warns of too high temperature and blue indicates too low temperature. The care-giver can accordingly add water to the sand to increase cooling, or lift the child away for a while if the temperature is too low.



Charge your devices using a cooking pot

When you are camping in the middle of nowhere, and want to charge your phone, you could draw out the PowerPot and start cooking a stew. Voila, your devices will get charged. PowerPot is a cooking utensil that can convert changing temperatures (of any water-based food inside it) into electricity. It can be used for wildlife tours, medical camps or rescue missions—any place where you would be cooking but do not have electricity. The basic model costs \$149 but you could upgrade it with adaptors, lights, batteries, water purifiers, light-weight stove and other accessories.

Inside: The PowerPot uses a thermoelectric differential transducer to harness the power of changing water temperatures within the pot. The more the temperature difference, the more electricity is produced. There is a built-in regulator, which outputs 5 V (USB standard) and up to 1000 mA of current, which is enough to charge smartphones, MP3 players, radio, GPS devices, lights, water-purifiers and other such devices. It is almost like charging off a normal outlet at home. Simply fill the pot with fluid, plug-in the regulating cable and close the handles to guide it away from the heat. Place the pot on any heat source and start cooking. Within seconds, a green indicator lights up and you can start charging devices. The pot is waterproof, fire-resistant and light-weight. It does not rely on sunlight, so it can be used any time. There are no moving parts or batteries, and since the thermoelectric technology is built into the bottom of the pot it can produce electricity from a wide variety of heat sources. The only condition is that water must be used for cooking, since it acts as a heat-sink to maintain the temperature difference required to produce thermal electricity.

